

## 7.4.36

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## Question:

The abscissa of two points **A** and **B** are roots of the equation  $x^2 + 2ax - b^2 = 0$  and their ordinates are roots of the equation  $x^2 + 2px - q^2 = 0$ . Find the equation and the radius of the circle with  $AB$  as diameter.

# Theoretical Solution

Define the Problem in Vector Form

The vector equation for a circle is given by:

$$\|\mathbf{x}\|^2 + 2\mathbf{u}^T \mathbf{x} + f = 0$$

where  $\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}$ , the center is  $\mathbf{c} = -\mathbf{u}$ , and  $f = \|\mathbf{u}\|^2 - r^2$ . Let the points be  $\mathbf{x}_A = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$  and  $\mathbf{x}_B = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$ .

Determine the Vector  $\mathbf{u}$

Since  $AB$  is the diameter, the center  $\mathbf{c}$  is the midpoint of  $\mathbf{x}_A$  and  $\mathbf{x}_B$ .

$$\mathbf{c} = \frac{\mathbf{x}_A + \mathbf{x}_B}{2} = \frac{1}{2} \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \end{pmatrix}$$

# Theoretical Solution

Using Vieta's formulas for the given quadratic equations:

- For  $x^2 + 2ax - b^2 = 0$ :

$$x_1 + x_2 = -2a$$

$$x_1 x_2 = -b^2$$

- For  $y^2 + 2py - q^2 = 0$ :

$$y_1 + y_2 = -2p$$

$$y_1 y_2 = -q^2$$

Substituting the sum of roots into the center vector  $\mathbf{c}$ :

$$\mathbf{c} = \frac{1}{2} \begin{pmatrix} -2a \\ -2p \end{pmatrix} = \begin{pmatrix} -a \\ -p \end{pmatrix}$$

Therefore, the vector  $\mathbf{u}$  is:

$$\mathbf{u} = -\mathbf{c} = - \begin{pmatrix} -a \\ -p \end{pmatrix} = \begin{pmatrix} a \\ p \end{pmatrix}$$

# Theoretical Solution

Determine the Radius  $r$  and Scalar  $f$

The squared diameter  $d^2$  is  $\|\mathbf{x}_A - \mathbf{x}_B\|^2$ .

$$d^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$$

We use the identity  $(m - n)^2 = (m + n)^2 - 4mn$ .

$$\begin{aligned}(x_1 - x_2)^2 &= (x_1 + x_2)^2 - 4(x_1 x_2) \\ &= (-2a)^2 - 4(-b^2) \\ &= 4a^2 + 4b^2 = 4(a^2 + b^2)\end{aligned}$$

$$\begin{aligned}(y_1 - y_2)^2 &= (y_1 + y_2)^2 - 4(y_1 y_2) \\ &= (-2p)^2 - 4(-q^2) \\ &= 4p^2 + 4q^2 = 4(p^2 + q^2)\end{aligned}$$

The squared diameter is:

$$d^2 = 4(a^2 + b^2) + 4(p^2 + q^2) = 4(a^2 + b^2 + p^2 + q^2)$$

# Theoretical Solution

The squared radius  $r^2$  is  $d^2/4$ :

$$r^2 = a^2 + b^2 + p^2 + q^2$$

Now, we find the scalar  $f = \|\mathbf{u}\| - r^2$ .

$$\|\mathbf{u}\| = \left\| \begin{pmatrix} a \\ p \end{pmatrix} \right\|^2 = a^2 + p^2$$

$$\begin{aligned} f &= (a^2 + p^2) - (a^2 + b^2 + p^2 + q^2) \\ &= a^2 + p^2 - a^2 - b^2 - p^2 - q^2 \\ &= -(b^2 + q^2) \end{aligned}$$

# Theoretical Solution

Final Equation and Radius Substitute  $\mathbf{u}$  and  $f$  into the circle's vector equation:

$$\|\mathbf{x}\|^2 + 2\mathbf{u}^T \mathbf{x} + f = 0$$

$$\left\| \begin{pmatrix} x \\ y \end{pmatrix} \right\|^2 + 2 \begin{pmatrix} a & p \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} - (b^2 + q^2) = 0$$

$$(x^2 + y^2) + 2(ax + py) - (b^2 + q^2) = 0$$

$$x^2 + y^2 + 2ax + 2py - b^2 - q^2 = 0$$

**Equation of the circle:**

$$x^2 + y^2 + 2ax + 2py - b^2 - q^2 = 0$$

**Radius of the circle:**

$$r = \sqrt{a^2 + b^2 + p^2 + q^2}$$

# python Code

```
import sympy

def solve_circle_problem():
    x, y, a, b, p, q = sympy.symbols('x y a b p q')
    x1, x2, y1, y2 = sympy.symbols('x1 x2 y1 y2')

    print("--- Problem Setup ---")
    print("Equation for x-coordinates:  $x^2 + 2ax - b^2 = 0$ ")
    print("Equation for y-coordinates:  $y^2 + 2py - q^2 = 0$ \n")
    )

    sum_x = -2 * a
    prod_x = -b**2

    sum_y = -2 * p
    prod_y = -q**2

    print("--- Applying Vieta's Formulas ---")
    print(f"Sum of x-roots (x1 + x2): {sum x}")
```



```
center_h = sympy.simplify(sum_x / 2)
center_k = sympy.simplify(sum_y / 2)
print(f"--- Circle Center (h, k) ---")
print(f"h = (x1 + x2) / 2 = {center_h}")
print(f"k = (y1 + y2) / 2 = {center_k}\n")

x_diff_sq = sum_x**2 - 4 * prod_x
y_diff_sq = sum_y**2 - 4 * prod_y

diameter_sq = x_diff_sq + y_diff_sq
radius_sq = sympy.simplify(diameter_sq / 4)
radius = sympy.sqrt(radius_sq)

print("--- Circle Radius (r) ---")
print(f"Squared diameter d^2 = (x1-x2)^2 + (y1-y2)^2 = {
    diameter_sq}")
print(f"Squared radius r^2 = d^2 / 4 = {radius_sq}")
print(f"Radius r = {radius}\n")
```

```
circle_eq_standard = (x - center_h)**2 + (y - center_k)**2 -  
    radius_sq  
  
circle_eq_general = sympy.expand(circle_eq_standard)  
  
print("--- Final Equation of the Circle ---")  
print(f"General Form: {circle_eq_general} = 0")  
  
if __name__ == "__main__":  
    solve_circle_problem()
```